VibeStream: Hybrid Music Recommendation Engine

VibeStream is an advanced music recommendation system designed to analyze song characteristics and user behavioral patterns to provide highly personalized music suggestions.

1. Introduction

The goal of this project is to bridge the gap between simple genre-based recommendations and deep acoustic analysis. VibeStream uses a hybrid approach, combining Content-Based Filtering (Acoustic Features) and Collaborative Filtering (User Behavior) to identify songs with similar 'vibes'.

2. System Architecture

The system is divided into two main phases: the Training Phase and the Inference (UI) Phase.

Training Phase: Handled by train.py. It cleans the Spotify dataset, normalizes features using MinMaxScaler, and creates a behavioral pivot matrix to simulate user interactions. The final output is a compact .pkl bundle.

Inference Phase: Handled by app.py. A Streamlit interface allows users to select a song, which then triggers a real-time Cosine Similarity calculation against the normalized acoustic vectors.

3. Methodology

The core of the recommendation engine is the Cosine Similarity metric. By representing each song as a vector of 11 acoustic features (Danceability, Energy, Key, Loudness, Mode, Speechiness, Acousticness, Instrumentalness, Liveness, Valence, Tempo), the system calculates the angle between vectors to determine similarity.

Hybrid Logic: The system supports three modes: Acoustic (pure vector similarity), Behavioral (trend-based), and Hybrid (a weighted blend of both).

4. Technical Stack

5. Conclusion

VibeStream demonstrates the effectiveness of hybrid recommendation systems. By combining hard audio data with behavioral simulation, the engine provides a more nuanced and 'human' music discovery experience compared to traditional filtered lists.